Mathematics for Economists – SOLUTIONS

Equations and Algebra

1. Factorise and solve the following quadratic equations:
   a. $x^2 + 5x + 6 = \boldsymbol{(x+2)(x+3)}$
   b. $x^2 - 4 = \boldsymbol{(x+2)(x-2)}$
   c. $x^2 + 6x + 9 = \boldsymbol{(x+3)^2}$
   d. $5x^2 + 13x + 6 = \boldsymbol{(5x+3)(x+2)}$
2. Solve the following linear simultaneous equations:
   a. $y = 5x + 6; y = -3x + 2$

$$5x + 6 = -3x + 2 \Rightarrow 8x = -4 \Rightarrow x = -\frac{1}{2}$$

$$y = -\frac{5}{2} + 6 = \frac{7}{2}$$

$$\boldsymbol{\left(x = -\frac{1}{2}, y = \frac{7}{2}\right)}$$

   b. $2y = 4x + 2; -\frac{1}{2}y = \frac{5}{2}x - 3$

$$2y = 4x + 2 \Longrightarrow y = 2x + 1$$

$$-\frac{1}{2}y = \frac{5}{2}x - 3 \Rightarrow y = -5x + 6$$

$$2x + 1 = -5x + 6 \Rightarrow 7x = 5 \Rightarrow x = \frac{5}{7}$$

$$y = \frac{10}{7} + 1 = \frac{17}{7}$$

$$\boldsymbol{\left(x = \frac{5}{7}, y = \frac{17}{7}\right)}$$

   c. $3x + 2y = z + 11; 2x + z = 7 + 3y; 5x + y - 2x = 12$

$$3x + 2y - z = 11$$
$$2x - 3y + z = 7$$
$$5x + y - 2z = 12$$

Use Gaussian elimination by adding first two equations:

$$5x - y = 18$$

Multiply the second equation by 2, then add to the third equation:

$$9x - 5y = 26$$

Solve for one of the variables:

$$y = 5x - 18 \Longrightarrow 9x - 5(5x - 18) = 26$$
$$16x = 64 \Rightarrow x = 4$$

By substitution:

$$y = 5(4) - 18 = 2$$

By substitution:

$$3(4) + 2(2) - z = 11 \Rightarrow 14 - z = 11 \Rightarrow z = 3$$
$$\boldsymbol{(x = 4, y = 2, z = 3)}$$

3. Solve the following quadratic simultaneous equations:
   a. $y = x + 3; y = x^2 + 5x - 2$

$$x + 3 = x^2 + 5x - 2$$
$$x^2 + 4x - 5 = 0$$
$$(x + 5)(x - 1)$$
$$x = -5; x = 1$$
$$y = -5 + 3 = -2$$
$$y = -1 + 3 = 2$$
$$(\boldsymbol{x = -5, y = -2}); (\boldsymbol{x = -1, y = 2})$$

   b. $y = x^2 + 5; y = -x^2 + 15$

$$x^2 + 5 = -x^2 + 15$$
$$2x^2 = 10 \Rightarrow x = \pm\sqrt{5}$$
$$y = 5 + 5 = 10$$
$$(\boldsymbol{x = \sqrt{5}, y = 10}); (\boldsymbol{x = -\sqrt{5}, y = 10})$$

   c. $x^2 + y^2 = 25; y = x + 5$
   By substitution of linear into quadratic:

$$x^2 + (x + 5)^2 = 25$$
$$2x^2 + 25x = 0$$
$$x(2x + 25) = 0$$
$$x = 0, x = -\frac{25}{2}$$
$$y = 0 + 5 = 5$$
$$y = -\frac{25}{2} + 5 = -\frac{15}{2}$$
$$(\boldsymbol{x = 0, y = 5}); (\boldsymbol{x = -\frac{25}{2}, y = -\frac{15}{2}})$$

4. Simplify the following expressions:
   a. $\frac{x^{-\frac{1}{3}}}{x^{-1}} = x^{-\frac{1}{3}-(-1)} = x^{-\frac{1}{3}+1} = \boldsymbol{x^{\frac{2}{3}}}$
   b. $x^2 x^{-\frac{1}{7}} x^{\frac{1}{2}} = x^{2+\left(-\frac{1}{7}\right)+\left(\frac{1}{2}\right)} = \boldsymbol{x^{\frac{33}{14}}}$
   c. $(2^x)^2 4^{x^{\frac{1}{2}} x^{\frac{1}{3}}} 2^{x^3} = 2^{2x} 4^{x^{\frac{5}{6}}} 2^{x^3} = 2^{2x} (2^2)^{x^{\frac{5}{6}}} 2^{x^3} = 2^{2x} 2^{2x^{\frac{5}{6}}} 2^{x^3} = \boldsymbol{2^{2x+2x^{\frac{5}{6}}+x^3}}$

5. (***Challenge***) Solve the following equations simultaneously:

$$y = x^{\frac{1}{2}} - 3; 5y = -x^{\frac{1}{2}} - 5$$
$$x^{\frac{1}{2}} = -\frac{1}{5}x^{\frac{1}{2}} + 2 \Longrightarrow 5x^{\frac{1}{2}} = -x^{\frac{1}{2}} + 10 \Rightarrow 6x^{\frac{1}{2}} = 10 \Rightarrow x^{\frac{1}{2}} = \frac{5}{3} \Longrightarrow x = \frac{25}{9}$$
$$y = \frac{5}{3} - 3 = -\frac{4}{3}$$
$$\left(\boldsymbol{x = \frac{25}{9}, y = -\frac{4}{3}}\right)$$

6. (***Challenge***) When using the Lagrangian Multiplier Method (*do not worry about what this is*), we often end up with a collection of equations that need to be solved simultaneously. Find *all* the solutions to the following (ignoring λ):

$$2x = \lambda(10x + 6y)$$
$$2y = \lambda(10y + 6x)$$

$$5x^2 + 6xy + 5y^2 = 1$$

Notice that we have three equations and three unknowns, hence a solution can be identified. First, we isolate the variable λ in the first equation:

$$\lambda = \frac{2x}{10x + 6y}$$

Substituting into the second equation yields:

$$2y = \frac{2x}{10x + 6y}(10y + 6x) \Rightarrow \frac{y}{x} = \frac{10y + 6x}{10x + 6y} \Rightarrow 6y^2 + 10xy = 10xy + 6x^2 \Longrightarrow x^2 = y^2$$

Substituting this result into the final equation implies:

$$10x^2 \pm 6x^2 = 1$$

The reason there is a plus-minus there is because $y = \pm x$ when we square root our previous result! Hence, we have a range of possibilities here:

$$10x^2 + 6x^2 = 16x^2 = 1 \Rightarrow x = \frac{1}{4}, -\frac{1}{4}$$

$$10x^2 - 6x^2 = 4x^2 = 1 \Rightarrow x = \frac{1}{2}, -\frac{1}{2}$$

Hence, there are eight solutions:

$$\boldsymbol{\left(x = \frac{1}{4}, y = \frac{1}{4}\right), \left(x = \frac{1}{4}, y = -\frac{1}{4}\right)}$$

$$\boldsymbol{\left(x = -\frac{1}{4}, y = \frac{1}{4}\right), \left(x = -\frac{1}{4}, y = -\frac{1}{4}\right)}$$

$$\boldsymbol{\left(x = \frac{1}{2}, y = \frac{1}{2}\right), \left(x = \frac{1}{2}, y = -\frac{1}{2}\right)}$$

$$\boldsymbol{\left(x = -\frac{1}{2}, y = \frac{1}{2}\right), \left(x = -\frac{1}{2}, y = -\frac{1}{2}\right)}$$

We ignore λ as required.

7. (***Challenge***) Solve the following simultaneous equations:

$$\frac{8^x}{128} = \frac{1}{4^y}; \frac{49^x}{7^y} = 1$$

$$49^x = 7^y \Rightarrow 7^{2x} = 7^y \Rightarrow 7^{2x-y} = 1$$

This equation can only be 1 if the power is equal to zero because only $7^0 = 1$. Hence, we know that:

$$2x - y = 0$$

For the other equation:

$$8^x 4^y = 128$$

$$2^{3x} 2^{2y} = 2^{3x+2y} = 128$$

Again, note that only $2^7 = 128$. Hence, we know that:

$$3x + 2y = 7$$

Now we have two linear simultaneous equations:

$$y = 2x \Longrightarrow 3x + 4x = 7 \Rightarrow x = 1 \Rightarrow y = 2$$

$$\boldsymbol{(x = 1, y = 2)}$$

Functions and Inverses

1. Solve the following absolute value equation: $|2x - 4| = 10$

$2x - 4 = \pm 10$
$2x - 4 = 10$ or $2x - 4 = -10$
$2x = 14$ or $2x = -6$
$x = 7$ or $x = -3$

2. Given the functions $f(x) = x^3 + 5x + 1$ and $g(x) = x^{\frac{1}{3}} + 2x^3$ state $f(g(x))$ and $g(f(x))$.

$$f(g(x)) = \left(\boldsymbol{x^{\frac{1}{3}} + 2x^3}\right)^{\boldsymbol{3}} + \mathbf{5}\left(\boldsymbol{x^{\frac{1}{3}} + 2x^3}\right) + \mathbf{1}$$
$$g(f(x)) = \boldsymbol{(x^3 + 5x + 1)^{\frac{1}{3}} + 2(x^3 + 5x + 1)^3}$$

3. State the degree of the following homogeneous functions:
   a. $F(x, y,) = x^{\frac{1}{3}}y^{\frac{2}{3}}$

$$F(\alpha x, \alpha y) = (\alpha x)^{\frac{1}{3}}(\alpha y)^{\frac{2}{3}} = \alpha^{\frac{1}{3}}\alpha^{\frac{2}{3}}F(x, y) = \alpha F(x, y)$$

   Hence, it is homogeneous of **degree 1.**

   b. $F(x, y) = \min\{x, y\}$

$$F(\alpha x, \alpha y) = \min\{\alpha x, \alpha y\} = \alpha \min\{x, y\} = \alpha F(x, y)$$

   Hence, it is homogeneous of **degree 1.**

   c. $F(x, y) = x^2y^2$

$$F(\alpha x, \alpha y) = (\alpha x)^2(\alpha y)^2 = \alpha^4x^2y^2 = \alpha^4F(x, y)$$

   Hence, it is homogeneous of **degree 4.**

   d. $F(x, y) = x^3 + y^3$

$$F(\alpha x, \alpha y) = (\alpha x)^3 + (\alpha y)^3 = \alpha^3x^3 + \alpha^3y^3 = \alpha^3F(x, y)$$

   Hence, it is homogeneous of **degree 3.**

4. Are the following functions monotonous? (*Hint:* Use *Desmos* to graph each one if unsure)
   a. $y = x^3$: **yes**
   b. $y = x^2 + 3$: **no**
   c. $y = 5x - 8$: **yes**
   d. $y = x^{\frac{1}{2}}$: **yes**
   e. $y = e^x$: **yes**
5. Find the inverse function of the following functions and verify your solutions using the inverse function identity:
   a. $f(x) = 5x + 7 \Rightarrow \boldsymbol{f^{-1}(x) = \frac{x-7}{5}}$; $\boldsymbol{f^{-1}(f(x)) = \frac{(5x+7)-7}{5} = x}$

b. $f(x) = -x + 3 \Rightarrow \boldsymbol{f^{-1}(x) = -x + 3};\ \boldsymbol{f^{-1}(f(x))} = \boldsymbol{-(-x+3) + 3 = x}$
c. $f(x) = x^{\frac{1}{3}} \Rightarrow \boldsymbol{f^{-1}(x) = x^3};\ \boldsymbol{f^{-1}(f(x)) = (x^{\frac{1}{3}})^3 = x}$

6. Does the function $f(x) = x^2$ have an inverse function? Why, or why not?

   Answer: Functions are rules that assign to some inputs one number as an output. In such a case, the quadratic function ***does not*** have an inverse. This is because for each y there are two values of x that could have produced it. For example, $y = 25$ could have been produced by $x = 5$ or $x = -5$. The quadratic could have an inverse only if we restricted its domain to only take positive numbers of x.

Limits

Evaluate the following limits:

1. $\lim_{x \to \infty} \frac{1}{x^2} = \mathbf{0}$

2. $\lim_{x \to 2} (x^2 - 4) = \mathbf{0}$

3. $\lim_{x \to 2} \frac{x+1}{x-1} = \mathbf{3}$

4. $\lim_{x \to 3} \frac{x-3}{x^2-5x+6} = \lim_{x \to 3} \frac{x-3}{(x-3)(x-2)} = \lim_{x \to 3} \frac{1}{x-2} = \mathbf{1}$

5. $\lim_{x \to 2} \frac{x^2-4}{x-2} = \lim_{x \to 2} \frac{(x-2)(x+2)}{x-2} = \lim_{x \to 2} x + 2 = \mathbf{4}$

Logarithms and Exponentials

1. Simplify the following logarithmic expressions:
   a. $\ln e^x = \boldsymbol{x}$
   b. $\ln(x^2 y) = (\mathbf{2ln}\,\boldsymbol{x} + \mathbf{ln}\,\boldsymbol{y})$
   c. $\ln\left(\frac{x^{-2}}{y}\right) = \mathbf{ln}\,\boldsymbol{y} - \mathbf{2}\,\mathbf{ln}\,\boldsymbol{x}$
   d. $\ln(x^2 e^{-x}) = \mathbf{2}\,\mathbf{ln}\,\boldsymbol{x} - \boldsymbol{x}$

2. It can be shown that an equation that is in log-log form is such that the coefficients are elasticities. What does this imply if for the income, cross-price, and own-price elasticity of demand for lemonade if the demand curve for lemonade is given by the following equation where p_l is the price of lemonade and p_c is the price of coke, and $Y$is income?

$$Q_d = Y p_l^{-\frac{1}{2}} p_c^{\frac{1}{3}}$$

   Answer: Take natural logarithm of both sides:

$$\boldsymbol{\ln Q_d = \ln Y - \frac{1}{2} \ln p_l + \frac{1}{3} \ln p_c}$$

   In terms of elasticities, this equation means that a 1% increase in income leads to a 1% increase in quantity demanded, a 1% increase in the price of lemonade leads to a 0.5%

decrease in quantity demanded, and a 1% increase in the price of coke leads to a 0.3% increase in the quantity demanded of lemonade.

Differentiation

1. Find the derivative of the following functions with respect to x:
   a. $f(x) = x^3$

      Apply exponent rule:

      $$\boldsymbol{\frac{df}{dx} = 3x^2}$$

   b. $f(x) = x^{-2}$

      Apply exponent rule:

      $$\boldsymbol{\frac{df}{dx} = -2x^{-3}}$$

   c. $f(x) = 5e^{x^2}$

      Apply chain rule:

      $$\boldsymbol{\frac{df}{dx} = 10xe^{x^2}}$$

   d. $f(x) = \ln(2x)$

      Apply chain rule:

      $$\boldsymbol{\frac{df}{dx} = \frac{1}{x}}$$

   e. $f(x) = \ln(x^3 + x^2 + 1)$

      Apply chain rule:

      $$\boldsymbol{\frac{df}{dx} = \frac{3x^2 + 2x}{x^3 + x^2 + 1}}$$

   f. $f(x) = \ln((x + 6)(e^x + 1))$

      First, let's multiply out this expression inside the logarithm:

      $$\ln((x + 6)(e^x + 1)) = \boldsymbol{\ln(xe^x + x + 6e^x + 6)}$$

      When differentiating, we apply both chain and product rules:

      $$\frac{df}{dx} = \frac{1}{e^x + x + 6e^x + 6} \times (e^x + xe^x + 6e^x + 1) = \boldsymbol{\frac{e^x(7 + x) + 1}{e^x + x + 6e^x + 6}}$$

g. $f(x) = e^x(x^2 + 1)$

Apply product rule:

$$\frac{df}{dx} = e^x(x^2 + 1) + 2e^x x = e^x(x^2 + 2x + 1) = \boldsymbol{e^x(x+1)^2}$$

h. $f(x) = \frac{x^{\frac{1}{2}}+6}{x+1}$

Apply quotient rule:

$$\boldsymbol{\frac{df}{dx} = \frac{\frac{1}{2}x^{-\frac{1}{2}}(x+1) - x^{\frac{1}{2}} + 6}{(x+1)^2}}$$

i. $f(x) = (x^2 \ln(x+5))^{-\frac{1}{6}}$

Apply chain rule multiple times and chain rule:

$$\boldsymbol{\frac{df}{dx} = -\frac{1}{6}(x^2 \ln(x+5))^{-\frac{7}{6}}\left(2x\ln(x+5) + \frac{x^2}{x+5}\right)}$$

j. $f(x) = 2^x$

Use the definition of a logarithm:

$$f(x) = 2^x = e^{\ln 2^x} = e^{x \ln 2}$$

Now using the chain rule, we see that:

$$\frac{df}{dx} = e^{x\ln 2}\ln 2 = \boldsymbol{2^x \ln 2}$$

2. Find the stationary points of the function $f(x) = \frac{5x^2+4}{x}$

To find the stationary points, we need to set $\frac{df}{dx} = 0$. We find the derivative either by quotient rule or by first simplifying the equation. The second approach is easier:

$$f(x) = 5x + 4x^{-1} \Rightarrow \frac{df}{dx} = 5 - 4x^{-2} = 0$$

$$5 = \frac{4}{x^2} \Longrightarrow 5x^2 = 4 \Rightarrow \boldsymbol{x} = \pm\sqrt{\boldsymbol{\frac{4}{5}}}$$

3. It is known that the growth of a variable over time is given by $\frac{dy}{dt}\frac{1}{y}$. Suppose that we know that $y = t^2 - t$. Work out the growth rate of y. Then, show that we can equivalently get the growth rate by taking logarithms of both sides and then differentiating with respect to time.

$$\frac{dy}{dt} = 2t - 1 \Rightarrow \frac{dy}{dt}\frac{1}{y} = \boldsymbol{\frac{2t-1}{y}}$$

By chain rule, when we take logarithms:

$$\frac{d\,(\ln y)}{dt} = \frac{1}{y} \times \frac{dy}{dt} = \boldsymbol{\frac{2t-1}{y}}$$

4. Find the second derivative for the function $y = e^{x-x^2}$

Apply chain rule:

$$\boldsymbol{\frac{dy}{dx} = (1-2x)e^{x-x^2}}$$

Apply chain and product rules:

$$\boldsymbol{\frac{d^2y}{dx^2} = -2e^{x-x^2} + (1-2x)^2 e^{x-x^2} = e^{x-x^2}\left((1-2x)^2 - 2\right)}$$

Partial Differentiation

Practice Problems

1. Find the partial derivative of each function with respect to x:
   a. $f(x,y) = ye^x$

$$\boldsymbol{\frac{\partial f}{\partial x} = ye^x}$$

   b. $f(x,y) = (x+y)^{\frac{1}{2}}$

$$\boldsymbol{\frac{\partial f}{\partial x} = \frac{1}{2}(x+y)^{-\frac{1}{2}}}$$

   c. $f(x,y) = x^2 + y^2$

$$\boldsymbol{\frac{\partial f}{\partial x} = 2x}$$

   d. $f(x,y) = 5xy$

$$\boldsymbol{\frac{\partial f}{\partial x} = 5y}$$

   e. $f(x,y) = \frac{y^2}{x}$

$$\frac{\partial f}{\partial x} = -y^2x^{-2} = -\frac{y^2}{x^2} = -\left(\frac{y}{x}\right)^2$$

f. $f(x,y) = \frac{x^2y^3}{\sqrt{x+y}}$

$$\frac{\partial f}{\partial x} = \frac{2xy^3\sqrt{x+y} - \frac{1}{2}x^2y^3(x+y)^{-\frac{1}{2}}}{(x+y)}$$

g. $f(x,y) = x^2ye^{xy}$

$$\frac{\partial f}{\partial x} = 2xye^{xy} + x^2y^2e^{xy} = e^{xy}(2xy + x^2y^2)$$

2. For the function $f(x,y) = x^2 + x + y^2 - 2$ find the stationary points by taking partial derivatives with respect to both variables and setting each expression equal to zero.

$$\frac{\partial f}{\partial x} = 2x + 1 = 0$$
$$\frac{\partial f}{\partial y} = 2y = 0$$
$$x = -\frac{1}{2}, y = 0$$

3. The Hessian matrix is a 2×2 table of second-order partial derivatives. Find this matrix for the function $f(x,y) = x^{\frac{1}{2}}y$ by finding $\frac{\partial^2 f}{\partial x^2}, \frac{\partial^2 f}{\partial y^2}, \frac{\partial^2 f}{\partial x \partial y}$ and $\frac{\partial^2}{\partial y \partial x}$.

   The two partial derivatives are:

$$\frac{\partial f}{\partial x} = \frac{1}{2}x^{-\frac{1}{2}}y$$
$$\frac{\partial f}{\partial y} = x^{\frac{1}{2}}$$

   We then take partial derivatives again:

$$\begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix} = \begin{bmatrix} -\frac{1}{4}x^{-\frac{3}{2}}y & \frac{1}{2}x^{-\frac{1}{2}} \\ \frac{1}{2}x^{-\frac{1}{2}} & 0 \end{bmatrix}$$

   Notice how the second-order mixed partial derivatives in the upper right and lower left corners are the same! This is almost always going to be true and is a result of Young's theorem.

Challenge Problems

1. For the function $f(x,y) = e^{y+x+1}$, find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$.

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y} = e^{y+x+1}$$

2. For the function $f(x,y) = \ln(x+y)$, find $\frac{\partial f}{\partial x}$ and $\frac{df}{\partial x \partial y}$.

$$\boldsymbol{\frac{\partial f}{\partial x} = \frac{1}{x+y}}$$

$$\boldsymbol{\frac{\partial f}{\partial x \partial y} = -\frac{1}{(x+y)^2}}$$

3. Young's theorem states that for any "well-defined" function, $\frac{df}{\partial x \partial y} = \frac{\partial f}{\partial y \partial x}$. Verify this theorem for the function $f(x,y) = x^{\frac{1}{3}} y^{\frac{2}{3}}$.

   The two partial derivatives are:

$$\frac{\partial f}{\partial x} = \frac{1}{3} x^{-\frac{2}{3}} y^{\frac{2}{3}}$$
$$\frac{\partial f}{\partial y} = \frac{2}{3} x^{\frac{1}{3}} y^{-\frac{1}{3}}$$

   We then take partial derivatives again:

$$\begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix} = \begin{bmatrix} \boldsymbol{-\frac{2}{9} x^{-\frac{5}{3}} y^{\frac{2}{3}}} & \boldsymbol{\frac{2}{9} x^{-\frac{2}{3}} y^{-\frac{1}{3}}} \\ \boldsymbol{\frac{2}{9} x^{-\frac{2}{3}} y^{-\frac{1}{3}}} & \boldsymbol{-\frac{2}{9} x^{\frac{1}{3}} y^{-\frac{5}{3}}} \end{bmatrix}$$

   We can see that indeed, $\frac{df}{\partial x \partial y} = \frac{\partial f}{\partial y \partial x}$, as $\boldsymbol{\frac{2}{9} x^{-\frac{2}{3}} y^{-\frac{1}{3}} = \frac{2}{9} x^{-\frac{2}{3}} y^{-\frac{1}{3}}}$, as required.

4. The chain rule for partial differentiation $\frac{df}{dt} = \frac{\partial f}{\partial t} + \frac{\partial f}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial t}$. Using this result, derive $\frac{df}{dt}$ for the function $f(x(t), y(t), t) = x^2 + 6y + t$, where $x(t) = t^2 - 8t$ and $y(t) = \frac{1}{t^2}$.

   The collection of necessary partial derivatives is:

$$\frac{\partial f}{\partial t} = 1$$

$$\frac{\partial f}{\partial x} = 2x$$

$$\frac{\partial x}{\partial t} = 2t - 8$$

$$\frac{\partial f}{\partial y} = 6$$

$$\frac{\partial y}{\partial t} = -2t^{-3}$$

   Putting them together yields:

$$\frac{df}{dt} = 1 + 2x(2t-8) - 6(2t^{-3}) = \boldsymbol{1 + 2(t^2 - 8t)(2t-8) - 12t^{-3}}$$

We can verify our answer by substitution:

$f(t) = (t^2 - 8t)^2 + 6\frac{1}{t^2} + t \Rightarrow \boldsymbol{\frac{df}{dt} = 2\left(t^2 - 8t\right)(2t - 8) - 12t^{-3}}$

We can see that the answers are the same both times!

5. Euler's theorem states that for a homogeneous function $f(x, y)$ of degree r, it holds that $\frac{\partial f}{\partial x}x + \frac{\partial f}{\partial y}y = rf(x, y)$. Using the chain rule for partial derivatives and the definition of a homogenous function, prove this result.

   By definition, we have that:

$$\alpha^r f(x, y) = f(\alpha x, \alpha y)$$

   Taking the derivative with respect to α yields:

$$r\alpha^{r-1} f(x, y) = \frac{\partial f}{\partial x}\frac{\partial x}{\partial a} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial a}$$

   Notice that since we are multiplying by a constant $\frac{\partial x}{\partial a} = x$ and $\frac{\partial y}{\partial a} = y$. Simplifying yields:

$$\alpha^{r-1} f(x, y) = \frac{\partial f}{\partial x}x + \frac{\partial f}{\partial y}y$$

   Notice that α is arbitrary, so we can simply state that $\alpha = 1$:

$$f(x, y) = \frac{\partial f}{\partial x}x + \frac{\partial f}{\partial y}y$$

   Thus, proving the theorem, as required.